

ChIA-PET Input Data

anchors, PET clusters, singletons, segments

Hierarchical Tree Construction

chromosome > segment > anchor > subanchor

CUDA

Stage 1: Heatmap Monte Carlo

Warp-level simulated annealing
Territory and segment positioning

CUDA

Stage 2: Arc-distance Monte Carlo

Warp-level simulated annealing
Anchor refinement from PET clusters

CUDA

Stage 3: Smoothing Monte Carlo

Polymer spring constraints
Subanchor bead chain generation

Quality Assessment

RMSD, Pearson r, stratum-adjusted correlation

3D Chromatin Structure

HCM, mmCIF, PDB output formats

Energy Function

distance
spring

Persistent GPU Resource Cache